Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/31/2005
PI Name:	Derby, Jeffrey J. Ph.D.		
Project Title:	First Principles Calculations of Molten II-IV Compounds and Their Solidification Behavior		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Electronic materials		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	55455-0132	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	05/15/2000	**End Date:**	11/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chelikowsky, J. (University of Minnesota)		
Grant/Contract No.:	NAG8-1676		
Performance Goal No.:			
Performance Goal Text:			

We have studied the properties of II-VI binary and ternary alloys using first-principles atomistic simulation techniques to better understand physical processes relevant to microgravity melt crystal growth. We have specifically studied:
II-VI binaries: We have studied the liquid state of CdTe and expanded our studies to encompass HgTe, ZnTe, and ZnSe. These choices were motivated by technological applications for optoelectronic substrates.

II-VI ternaries: Due to their significant technological importance, we have also studied the melt phase behavior of the ternary alloys of CdZnTe. We have studied several compositions, again motivated by technological uses.

Of particular interest are melt structure and physical properties that may impact the growth of single-crystal substrates from the melt. In addition to probing the physical behavior of these systems, our techniques have provided good predictions for many thermophysical properties, such as optical properties, electrical conductivities, specific heats, thermal conductivities, viscosities, diffusion coefficients.

Task Description:	The goal of our ground-based research project is to employ first-principles atomistic simulations to obtain a more fundamental understanding of the physical behavior of molten binary and ternary alloys of the group II elements cadmium (Cd), mercury (Hg), and zinc (Zn), and group VI elements tellurium (Te) and selenium (Se). Toward these ends, we will employ atomistic simulations with fully quantum mechanical forces of these systems. This procedure allows a parameter-free description of the liquid state and has been applied to liquid silicon, germanium and gallium arsenide; the resulting descriptions are in good agreement with existing experimental data. Other than our previous studies of liquid CdTe, no comparable simulations have been performed for II-VI semiconductors. The binary and ternary alloys of II-VI components are employed in a variety of technologically important electronic and electro-optical devices. Of special importance are room-temperature gamma ray detectors employing CdZnTe, which may permit a new generation of portable, sensitive radiation detectors to counter nuclear proliferation activities and terrorist threats such as dirty radiation bombs. To make these devices practical, a much better understanding of the materials processing of II-VI compounds is needed. The growth of high-quality, single-crystal substrates of these materials has typically proven to be extremely difficult under terrestrial conditions. As such, they present ideal model systems to probe using microgravity techniques. Indeed, many ongoing, NASA-funded space-flight and ground-based experiments are studying several technologically significant and scientifically interesting II-VI alloys. For example, growth experiments of CdZnTe in a microgravity environment aboard USML-1 resulted in material which was far superior in structural perfection compared to earth-grown material under similar conditions. We seek to obtain a more fundamental understanding of the properties of these molten liquids so that the physical mechanisms acting during crystal growth can be elucidated. An additional significant outcome of this work will be the theoretical prediction of valuable physical properties, such as diffusivities, which are not currently known and very difficult to measure accurately via experiments. Such predictions will be extremely useful for use in macroscopic process models and for comparison with NASA-funded experimental property measurements of II-VI compounds.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have completed our studies of transport properties of non-stoichiometric liquids of cadmium telluride. These studies have included computations of viscosity and self-diffusion coefficients at several compositions and temperatures. The results have been presented at several scientific conferences, and several publications are being written.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Articles in Peer-reviewed Journals	Ko E, Alemany MMG, Chelikowsky JR. "Viscosities of liquid CdTe near melting point from ab initio molecular-dynamics calculations." Journal of Chemical Physics. 2004 Jul;121(2):942. http://dx.doi.org/10.1063/1.1759612 , Jul-2004
Articles in Peer-reviewed Journals	Ko E, Alemany M, Derby JJ, Chelikowsky JR. "Ab Initio simulations of nonstoichiometric CdxTe1-x liquids." Journal of Chemical Physics, 2005 Aug;123(8):084508. http://dx.doi.org/10.1063/1.2008247 , Aug-2005
Books/Book Chapters	Derby JJ, Yeckel A. "Modeling of Crystal Growth Processes." in "Crystal Growth -- From Fundamentals to Technology." Ed. G. Muller, J.J. Metois, P. Rudolph. Amsterdam : Elsevier, 2004. p. 143-167., Jul-2004
Dissertations and Theses	Ko E. "First Principles Calculations of Semiconductor Liquids and Clusters." Dissertation, University of Minnesota, March 2004. , Mar-2004
Presentation	Ko, E. ; Alemany, M. ; Derby, J.J;. Chelikowsky, J.R. "Ab initio modeling of diffusion in off-stoichiometric semiconductor liquids" Modeling and Simulating Materials Nanoworld, 3rd International Conference on Computational Modeling and Simulation of Materials, Acireale, Sicily, Italy Jun-2004
Presentation	Derby, J.J. "Modeling of Crystal Growth Processes" ISSCG-12, The Twelfth International Summer School on Crystal Growth, Berlin, Germany Aug-2004
Presentation	Chelikowsky, J.R. "Atoms on the Move: Simulating the Properties of Semiconductor Liquids" Conference on Computational and Experimental Challenges, Los Angeles, CA Aug-2004
Presentation	Derby, J.J. "A Broader Overview of the Role of Modeling in the Development of Growth Methods for CdZnTe" The 2004 U.S. Workshop on the Physics and Chemistry of II-VI Materials, Chicago, IL Oct-2004